

CMOS Inverter Layout Design Using Magic VLSI Tool

VLSI Internship Project Report

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Duration: 6 Weeks

Abstract

This project presents the complete design, implementation, and verification of a CMOS inverter using Magic VLSI Tool with SCMOS technology. The CMOS inverter is one of the fundamental building blocks of digital VLSI circuits and is widely used in modern integrated circuit design due to its low power consumption, high noise immunity, and fast switching characteristics.

In this work, the inverter is designed at the layout level by carefully following SCMOS fabrication design rules. The physical layout is created using different layers such as diffusion, polysilicon, metal, and contacts to form PMOS and NMOS transistors in a complementary configuration. The layout is then verified using the Design Rule Check (DRC) tool to ensure compliance with fabrication constraints such as spacing, width, and enclosure rules.

After successful verification, the layout is processed to generate the SPICE netlist and extraction files, which are used for circuit-level simulation in NGSpice. A pulse input signal is applied to analyze the transient response of the circuit. The output waveform is observed to be the exact logical complement of the input waveform, confirming correct inverter operation.

This project demonstrates a complete VLSI design flow starting from layout creation, physical verification, extraction, and circuit simulation. It provides practical exposure to industry-standard tools and techniques used in CMOS digital design and forms a strong foundation for advanced VLSI system design.

1 Introduction

The rapid advancement of semiconductor technology has led to the development of highly integrated and low-power digital systems. Complementary Metal Oxide Semiconductor (CMOS) technology plays a crucial role in modern VLSI (Very Large Scale Integration) design due to its advantages such as low static power dissipation, high noise immunity, and high switching speed.

The CMOS inverter is the most fundamental and widely used digital logic gate in VLSI circuits. It serves as the basic building block for all complex digital systems including arithmetic circuits, memory elements, microprocessors, and communication systems. Because of its simplicity and importance, the CMOS inverter is often used as an introductory circuit in VLSI design flow.

A CMOS inverter consists of one PMOS transistor connected to the supply voltage (VDD) and one NMOS transistor connected to ground (GND). Both transistors share a common gate input. The output is obtained from the connected drains of both transistors. When the input is logic LOW, the PMOS transistor turns ON and the NMOS transistor turns OFF, resulting in a logic HIGH output. Similarly, when the input is logic HIGH, the NMOS transistor turns ON and the PMOS transistor turns OFF, producing a logic LOW output. This complementary operation ensures very low static power consumption.

In this project, the CMOS inverter is implemented using Magic VLSI Tool based on SCMOS technology rules. The design includes layout creation, verification using Design Rule Check (DRC), extraction of SPICE netlist, and circuit simulation using NGSpice. The objective is to understand the complete physical design flow and analyze the behavior of the inverter at both layout and circuit levels.

This work provides hands-on experience in VLSI design methodologies and demonstrates how theoretical concepts are translated into practical integrated circuit implementation using industry-standard tools.

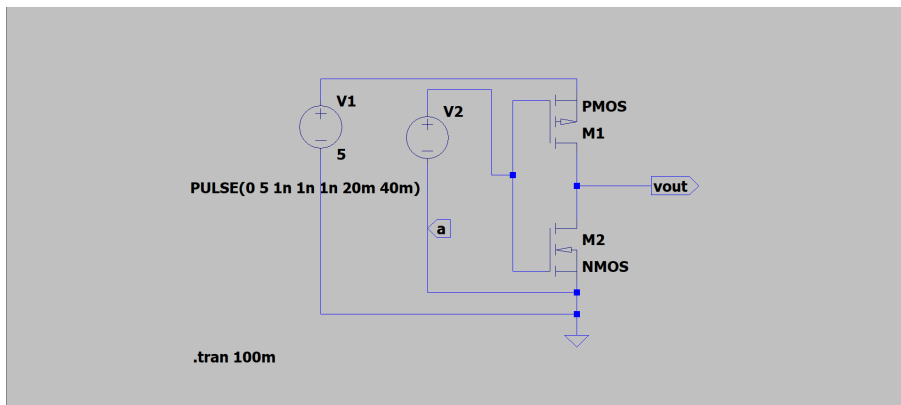


Figure 1: CMOS Inverter Schematic

2 Tools Used

- Magic VLSI Tool (Layout Design)
- SCMOS Technology Rules
- NGSpice (Circuit Simulation)

3 Objectives

The main objectives of this project are:

- To design a CMOS inverter layout using Magic VLSI Tool based on SCMOS technology rules.
- To understand the physical design flow of VLSI circuits from schematic to layout.
- To perform Design Rule Check (DRC) to ensure fabrication compatibility.
- To generate SPICE netlist and extraction files from the layout.
- To simulate the circuit using NGSpice and analyze transient response.
- To verify correct inverter operation by observing input and output waveforms.
- To gain practical exposure to VLSI physical design and simulation flow.

4 Layout Design

The CMOS inverter layout is designed using Magic VLSI Tool based on SCMOS fabrication rules. Layout design is the physical representation of the circuit where transistors and interconnections are implemented using different layers such as diffusion, polysilicon, metal, and contacts.

The design consists of two complementary transistors: a PMOS and an NMOS. The PMOS transistor is placed inside the n-well region and is connected to the supply voltage (VDD), while the NMOS transistor is placed in the p-substrate and connected to ground (GND). Both transistors share a common gate terminal, which acts as the input node of the inverter.

Polysilicon is used to form the gate terminal of both transistors, and diffusion regions are used to form the source and drain terminals. Metal layers are used for routing and interconnection between different nodes such as VDD, GND, input, and output. Contacts are placed between diffusion and metal layers to ensure proper electrical connectivity.

Careful attention is given to transistor sizing (W/L ratio) to achieve balanced switching characteristics and improve noise margins. The layout is optimized to reduce parasitic capacitance and area while maintaining correct functionality.

Proper spacing between layers is maintained according to SCMOS design rules to avoid short circuits and fabrication errors. After completing the layout, it is verified using Design Rule Check (DRC) to ensure that all physical design constraints are satisfied.

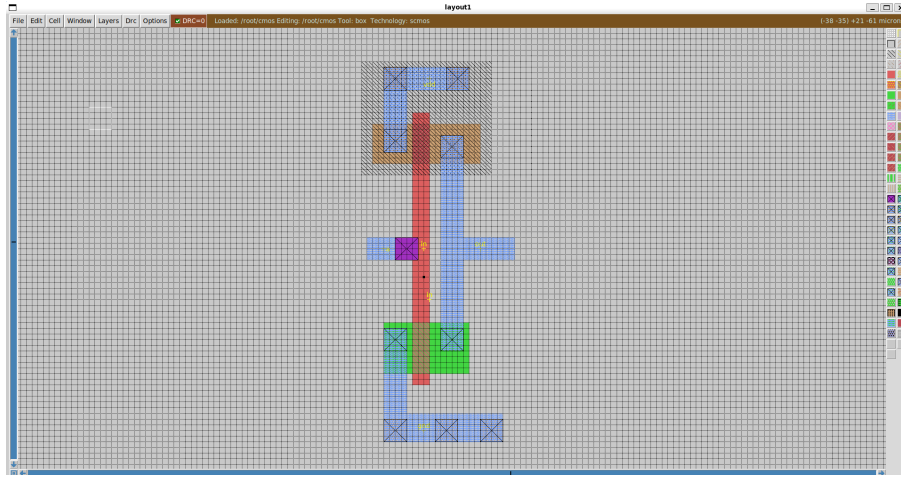


Figure 2: CMOS Inverter Final Layout

5 Design Rule Check (DRC)

Design Rule Check (DRC) is a critical verification step in the VLSI physical design flow that ensures the layout complies with the fabrication constraints defined by the semiconductor technology (SCMOS in this project). These rules are essential because any violation in spacing, width, or layer overlap can lead to manufacturing defects or circuit failure.

In this project, the CMOS inverter layout designed using Magic VLSI Tool was verified using the built-in DRC engine. The tool automatically checks all geometric constraints such as minimum width of diffusion and metal layers, minimum spacing between adjacent layers, proper overlap of contacts, and correct enclosure of vias and interconnections.

The DRC process helps in identifying errors like: - Short circuits between adjacent layers - Insufficient spacing between diffusion regions - Incorrect polysilicon gate overlap - Metal width violations - Improper contact placement

After running the DRC verification, the layout was found to be completely error-free with zero violations. This confirms that the design strictly follows SCMOS fabrication rules and is suitable for manufacturing. A clean DRC result is an important milestone in VLSI design as it ensures the reliability and correctness of the physical layout before extraction and simulation stages.

The successful DRC verification confirms that the CMOS inverter layout is fabrication-ready and meets all required design constraints defined by the technology node.

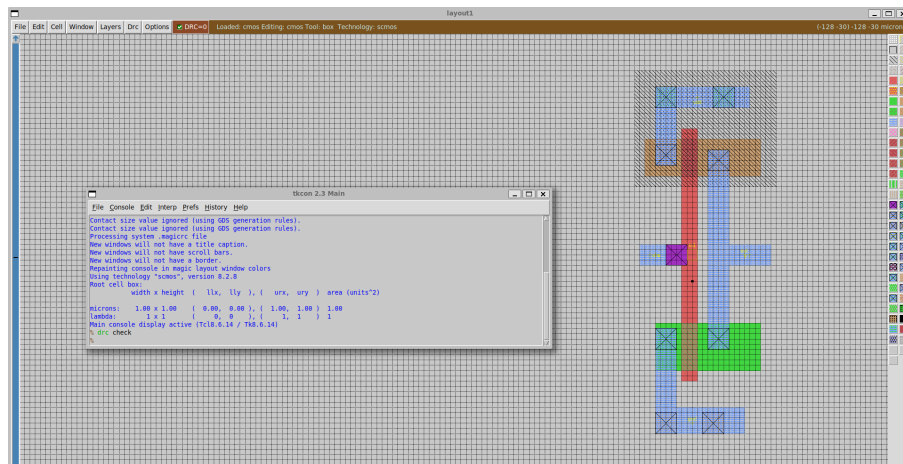


Figure 3: DRC Clean Verification Result

6 Generated Files

During the design and extraction process, several important files are generated from the layout. These files are used for simulation and further analysis.

CMOS.mag File: The .mag file is the primary layout database file used in Magic VLSI Tool. It stores all geometric information of the layout including layers, connections, and device placement.

CMOS.ext File: The .ext file contains extracted parasitic information such as resistance and capacitance values from the layout. This file is important for post-layout analysis and accurate simulation.

CMOS.spice File: The .spice file contains the transistor-level netlist of the CMOS inverter. It is used for circuit simulation in NGSpice to analyze dynamic behavior.

* CMOS Inverter SPICE Netlist

```
.option scale=1u
```

```
M1 y a vdd vdd pfet w=7 l=3
```

```
M2 y a gnd gnd nfet w=9 l=3
```

```
VDD vdd 0 DC 5
```

```
VA a 0 PULSE(0 5 0 1n 1n 10n 20n)
```

```
.model nfet nmos level=1
```

```
.model pfet pmos level=1
```

```

.tran 1n 100n
.end

* CMOS Inverter SPICE Netlist

.option scale=1u

M1 y a vdd vdd pfet w=7 l=3
M2 y a gnd gnd nfet w=9 l=3

VDD vdd 0 DC 5
VA a 0 PULSE(0 5 0 1n 1n 10n 20n)

.model nfet nmos level=1
.model pfet pmos level=1

.tran 1n 100n
.end

```

7 Waveform Analysis

The waveform analysis of the CMOS inverter is performed using NGSpice transient simulation. A pulse input signal is applied to the inverter to study its dynamic switching behavior and verify its logical operation.

Input Waveform

The input waveform is a pulse signal that alternates between logic LOW (0V) and logic HIGH (5V). This signal is applied to the gate terminals of both PMOS and NMOS transistors. The input waveform acts as the control signal that determines the switching behavior of the inverter. The rising and falling edges of the input are used to observe the transient response of the circuit.

Output Waveform

The output waveform is observed at the drain connection of the PMOS and NMOS transistors. The output is a digital signal that switches between logic HIGH and logic LOW depending on the input. When the input is LOW, the output becomes HIGH due

to the conduction of the PMOS transistor. When the input is HIGH, the output becomes LOW due to the conduction of the NMOS transistor. This confirms the correct inversion operation of the CMOS circuit.

Input and Output Waveform Comparison

The comparison of input and output waveforms clearly shows that the output is the logical complement of the input signal. When the input transitions from LOW to HIGH, the output transitions from HIGH to LOW, and vice versa. This opposite switching behavior confirms the correct functioning of the CMOS inverter.

The waveform also demonstrates proper switching characteristics without distortion, indicating correct transistor operation and successful circuit simulation using NGSpice.

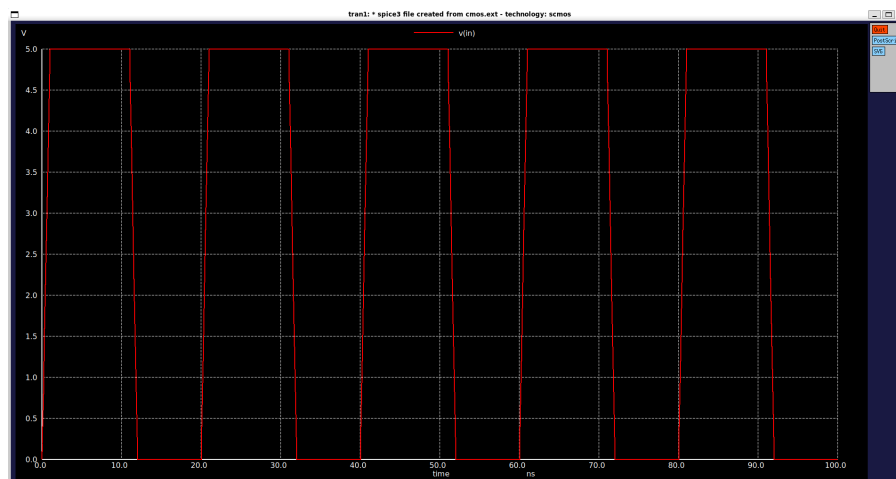


Figure 4: Input Waveform

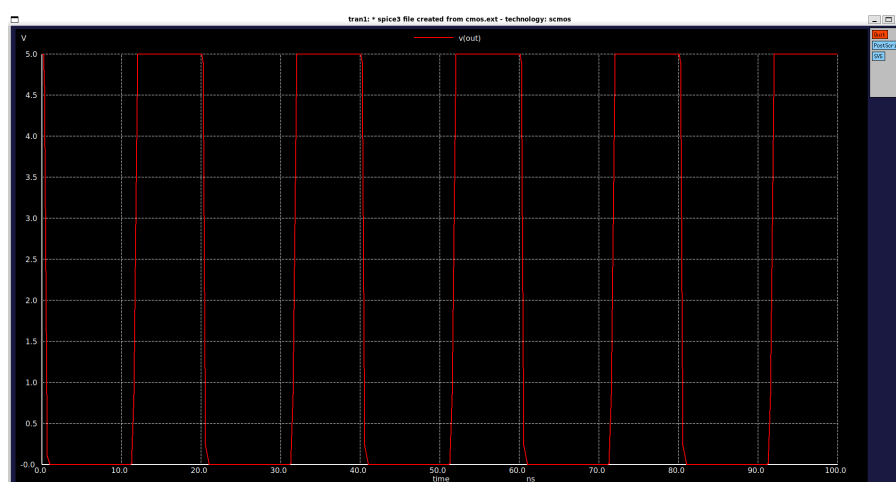


Figure 5: Output Waveform

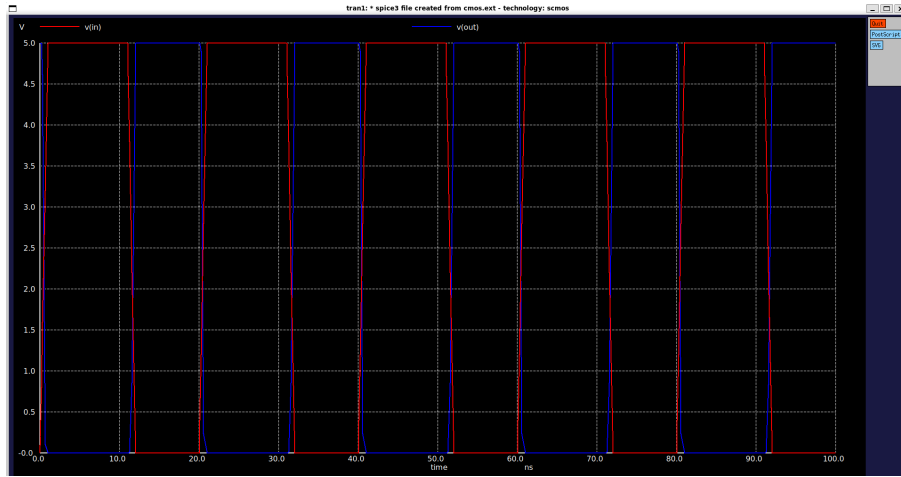


Figure 6: Input vs Output Waveform Comparison

8 Results

The CMOS inverter layout was successfully designed and verified using Magic VLSI Tool. The Design Rule Check was completed without any violations, confirming fabrication readiness. The SPICE netlist was extracted successfully and used for circuit simulation in NGSpice.

The simulation results clearly show that the output waveform is the logical complement of the input waveform, confirming correct inverter operation. This validates the entire VLSI design flow from layout creation to circuit-level verification.

9 Conclusion

The CMOS inverter was successfully designed, verified, and simulated using Magic VLSI Tool and NGSpice. The project demonstrates a complete VLSI physical design flow including layout design, design rule checking, netlist extraction, and circuit simulation.

The results confirm correct inverter functionality and demonstrate the importance of CMOS technology in modern digital circuits. This project provides strong practical understanding of VLSI design tools and methodology, forming a foundation for advanced circuit design projects.

10 Motivation

The motivation behind this project is to gain practical knowledge of VLSI physical design and understand how theoretical CMOS inverter concepts are implemented in real-world

semiconductor design.

Working with Magic VLSI Tool helps in understanding layout-level design, fabrication rules, and verification techniques. This project also builds interest in semiconductor technology and encourages further learning in advanced VLSI topics such as logic gate design, timing analysis, and processor architecture.

With the increasing demand for VLSI engineers in semiconductor industries, this project serves as a strong foundation for future career development.

11 Advantages

The CMOS inverter has several important advantages:

- Low static power consumption due to complementary operation of PMOS and NMOS transistors.
- High switching speed, suitable for digital applications.
- High noise immunity, ensuring stable operation in noisy environments.
- Simple structure, making it easy to design and analyze.
- Scalable design, allowing integration into complex digital circuits.
- Widely used as a fundamental building block in VLSI systems.

12 Applications

The CMOS inverter is widely used in various digital and electronic systems:

- Design of logic gates such as NAND, NOR, XOR, and XNOR.
- Microprocessors and microcontrollers.
- Memory circuits such as SRAM and DRAM.
- Digital signal processing systems.
- Integrated circuit design in VLSI technology.
- Used as a basic building block in complex digital systems.

13 Future Scope

This project can be extended to design more complex digital circuits such as NAND, NOR, XOR, and full adder circuits using CMOS technology.

Future improvements may include Layout Versus Schematic (LVS) verification, parasitic extraction analysis, timing analysis, and power optimization techniques. Advanced research can focus on deep submicron technologies and nanoscale CMOS design.

This work can also be extended to build complete digital subsystems such as arithmetic logic units (ALU), multipliers, and microprocessor cores, which are widely used in modern semiconductor industries.

14 References

- Magic VLSI Tool Documentation
- NGSpice User Manual
- SCMOS Design Rules Guide
- CMOS VLSI Design - Weste & Harris